

AI- and MBSE-Based Agile Satellite Design

CHAPTER 5

Decomposition of Digital Satellite Design Indicators

5.1 Analysis of Satellite Functions and Indicators

Within the MBSE framework, satellite functional modeling and indicator modeling provide the essential support for moving from a system concept to a verifiable design. Together, they constitute the core pillars of digital design. They not only support the systematic transformation of requirements into engineering implementation, but also use formalized methods to map complex aerospace mission objectives accurately into a network of decomposable, traceable, and verifiable functional structures and technical indicators. As satellites are iterated rapidly for scientific exploration, constellation networking, and commercial missions, the industry has placed higher demands on the collaborative efficiency, engineering accuracy, and reusability of integrated model-driven functional and indicator modeling.